Alagille Syndrome, Part Two: The Clinical Criteria. Five Classic Features, the Kidney and the Vessels, and Modified Criteria When the Variant Is Known.

Imagine a baby girl, seven weeks old, who is still jaundiced. Her conjugated bilirubin is four point nine milligrams per decilitre, and her gamma-glutamyl transferase, the enzyme that rises when bile ducts are injured, is four hundred and eighty units per litre. A hepatobiliary scan, in which a radioactive tracer is injected and the camera waits to see it appear in the intestine, shows no excretion at all after twenty-four hours. A liver biopsy has been done. The pathologist counted eleven portal tracts and found bile ducts in them, together with a ductular reaction, which means a proliferation of small reactive ductules at the edge of the portal tracts. There is no paucity of bile ducts. The surgical team has booked an intraoperative cholangiogram for the next morning, with a plan to proceed to a Kasai portoenterostomy if the extrahepatic bile ducts look atretic.

There is one more thing. On examination there is a soft systolic murmur that can be heard in both armpits, and an echocardiogram shows mild stenosis of the branch pulmonary arteries. Later, an ophthalmologist examines the mother, who is completely well, and finds a posterior embryotoxon in one eye.

Five questions follow. Does this infant have Alagille syndrome, given that she has only two of the five classic features and no paucity on biopsy? Does a normal duct count at seven weeks argue against the diagnosis? Should the operation go ahead, or should she first have a spine film, an eye examination and a gene panel? Does the finding in the mother count as a family history? And if a variant in the gene JAG1 is found, which other organs must be examined, and when?

The main idea of this chapter is that the classic diagnostic rule for Alagille syndrome, bile duct paucity on biopsy plus at least three of five clinical features, was built from children who were already obviously affected. It works well for them. It works badly in three situations that matter a great deal in practice: in young infants, whose livers may not yet show paucity; in relatives, who carry the same variant but often have only one mild feature or none; and in patients with variants in NOTCH2, the second gene, who often lack the classic face, eye and spine findings. The modern diagnosis rests on genotype plus any single feature, and the evaluation must extend to the kidney and the blood vessels of the brain, which the classic five features ignore even though they cause a substantial share of illness and death.

A few terms first. Bile duct paucity, also called ductopenia, means too few interlobular bile ducts in the portal tracts on biopsy. It is usually defined as a ratio of bile ducts to portal tracts below zero point nine, where the normal ratio lies between zero point nine and one point eight, and pathologists generally want at least ten portal tracts to make the judgement. Posterior embryotoxon is a prominent, forward-displaced Schwalbe line, the ring at the edge of the cornea where it meets the iris angle, seen with a slit lamp. Butterfly vertebrae are vertebral bodies that

failed to fuse in the midline during development, so that on a frontal radiograph each one looks like a pair of wings. Peripheral pulmonary stenosis means narrowing of the branch pulmonary arteries beyond the main pulmonary trunk. Penetrance is the proportion of carriers who show any sign of a condition, and expressivity describes how severe those signs are.

The evidence in this chapter comes from several sources. The original descriptions are by Daniel Alagille and colleagues in the Journal of Pediatrics in 1975 and 1987. The most widely quoted feature frequencies come from Emerick and colleagues at the Children's Hospital of Philadelphia, who described ninety-two patients in Hepatology in 1999. Cardiac data come from McElhinney and colleagues in Circulation in 2002, facial features from Kamath and colleagues in the American Journal of Medical Genetics in 2002, family studies from Kamath and colleagues in the Journal of Medical Genetics in 2003, vascular disease from Kamath and colleagues in Circulation in 2004 and from Emerick and colleagues in the Journal of Pediatric Gastroenterology and Nutrition in 2005, the effect of the Kasai operation from Kaye and colleagues in the same journal in 2010, and renal disease from Kamath and colleagues in the American Journal of Medical Genetics in 2012. Large international data come from the Global Alagille Alliance, known as GALA, reported by Vandriel and colleagues in Hepatology in 2023 and in Liver International in 2025. Guidance comes from the 2017 joint guideline on cholestatic jaundice in infants from the North American and European societies for paediatric gastroenterology, hepatology and nutrition, from the 2024 Clinical Practice Guidelines of the European Association for the Study of the Liver on genetic cholestatic liver diseases, and from the GeneReviews chapter on Alagille syndrome, last updated in January 2024. For brevity these will be called the 2017 joint guideline, the European guideline and GeneReviews.

Part one. Where the criteria came from

In 1975 Alagille and his colleagues in Paris described a group of children with chronic cholestasis who also had narrowing of the peripheral pulmonary arteries, defects of the vertebral arches, a characteristic face and delayed growth. They called it arteriohepatic dysplasia. In 1987 the same group reported eighty patients, and from that experience came the definition that is still taught today: paucity of interlobular bile ducts on liver biopsy, together with at least three of five major features, namely chronic cholestasis, a cardiac defect, skeletal abnormalities, ocular abnormalities and a characteristic face.

Ten years later, in 1997, two groups identified JAG1 as the gene responsible. From that moment the syndrome could be defined by genotype as well as by appearance, and family studies began to reveal relatives who carried the same variant but looked almost normal. In 2006 a second gene, NOTCH2, was added. By 2012 authors were arguing that renal involvement should be counted as a sixth defining feature, that vascular disease could reasonably be a seventh, and that the presence of cholestasis should be enough to satisfy the liver criterion without routine biopsy.

Why these particular organs? The answer is biological. JAG1 makes a signalling protein called Jagged1 that sits on the surface of one cell and activates the NOTCH2 receptor on a

neighbouring cell. This contact signal is used wherever the embryo needs to build a tube or a boundary: the bile ducts that form around the portal vein, the walls of the pulmonary arteries, the fusion of the two halves of each vertebral body, the angle at the front of the eye, and the mesenchyme of the face. The same pathway also builds kidney tubules and the walls of arteries elsewhere in the body. That is why renal and vascular disease belong to the syndrome, even though the 1987 rule left them out.

Part two. How often each feature occurs

The Emerick series of ninety-two children remains the reference for feature frequencies, and it is worth hearing the numbers one by one, remembering that not every child was examined for every feature. A cardiac murmur was present in ninety-seven percent, ninety of ninety-two children. Cholestasis was present in ninety-six percent. A characteristic face was described in ninety-six percent. Growth retardation was found in eighty-seven percent of the thirty-one children with adequate growth data. Bile duct paucity was present on biopsy in eighty-five percent, sixty-nine of eighty-one. Posterior embryotoxon was found in seventy-eight percent of those examined, and butterfly vertebrae in fifty-one percent of those who had radiographs. Renal disease was present in forty percent of the sixty-nine children evaluated, and fourteen percent, thirteen of ninety-two, had an intracranial bleed or stroke.

Three things should be taken from these figures. First, these were probands, the index patients, referred to a specialist centre because they were ill. The frequencies are the ceiling, not the average for everyone who carries the gene. Second, among the classic features, the eye and the spine are the least consistent, and the kidney is involved about as often as the spine, yet it is not in the rule. Third, the series also looked at prognosis. Only one clinical feature predicted death, and it was intracardiac congenital heart disease, that is, structural disease inside the heart such as tetralogy of Fallot, as opposed to narrowing of the peripheral pulmonary arteries alone. The predicted twenty-year survival was forty percent with intracardiac disease and eighty percent without it. Overall, twenty-one percent of the children needed a liver transplant, seventeen percent had died by the time of the report, and the predicted twenty-year survival for the whole group was seventy-five percent.

Part three. Why the liver criterion fails in young infants

The classic rule demands paucity on biopsy, but paucity develops over time. In the Emerick series there were eighty-eight biopsies from fifty-seven patients. Of biopsies taken before six months of age, only sixty percent, twenty-nine of forty-eight, showed paucity. Of biopsies taken after six months, ninety-five percent, thirty-eight of forty, did. The difference was highly significant. Put simply, a biopsy in the first months of life misses the diagnosis about two times in five.

Early biopsies can also actively mislead. Instead of too few ducts, they may show a ductular reaction or giant cell transformation of hepatocytes, the same pattern that is seen in biliary

atresia. Pathologists have described this as an evolution from proliferation to paucity over the first months of life.

The hepatobiliary scan misleads in the same direction. Among thirty-six infants with Alagille syndrome who had scintigraphy in the Emerick series, only five, fourteen percent, showed normal excretion within four hours. Nine had delayed excretion that appeared by twenty-four hours, and twenty-two, sixty-one percent, showed no excretion at all at twenty-four hours, which is exactly the result that raises concern for biliary atresia.

The European guideline reflects this. It states that the relative insensitivity of bile duct paucity in young infants with Alagille syndrome is well documented, recommends against routine liver biopsy for determining prognosis or management, with level of evidence three and a strong recommendation, and keeps a diagnostic role for biopsy mainly in limited situations, such as when genetic testing returns a variant of uncertain significance.

Part four. Why a Kasai operation in Alagille syndrome does harm

When an infant with Alagille syndrome has high-GGT cholestasis, a duct count that looks normal, a scan that shows no excretion and an intraoperative cholangiogram that shows a thin, poorly filled extrahepatic tree, the surgeon may conclude that this is biliary atresia and perform a Kasai portoenterostomy. In Alagille syndrome the extrahepatic ducts are often hypoplastic but patent. Removing them and connecting a loop of bowel to the liver hilum gains nothing, and it opens the way to ascending cholangitis.

Kaye and colleagues in Philadelphia compared children with Alagille syndrome who had undergone a Kasai procedure in infancy with those who had not. Liver transplantation was needed in forty-seven point three percent of the Kasai group, against thirteen point nine percent of the others. Death occurred in thirty-one point six percent of the Kasai group, against two point eight percent. A small Korean single-centre series by Lee and colleagues in 2015 pointed the same way: all five children who had had a Kasai went on to transplant, compared with two of ten who had not, and three of five died, compared with one of ten.

These are retrospective series, and there is real confounding: the infants sent for a Kasai were probably the sickest, so the operation may mark severity as much as cause harm. But there is no evidence that portoenterostomy helps, and every series points in the same direction. The 2017 joint guideline advises looking for features of Alagille syndrome before an intraoperative cholangiogram. In practice that means an echocardiogram, a frontal spine or chest film, a slit-lamp eye examination and a cholestasis gene panel while the infant is being worked up.

Part five. The heart

The heart is the most sensitive extrahepatic feature and the one that most influences survival. McElhinney and colleagues described two hundred individuals with a JAG1 variant or a clinical diagnosis of Alagille syndrome. Ninety-four percent, one hundred and eighty-seven of two

hundred, had some cardiovascular involvement. In seventy-five percent the anomaly was confirmed by imaging, and in a further nineteen percent the only evidence was a murmur, with a normal echocardiogram or no imaging. Stenosis or hypoplasia of the branch pulmonary arteries, documented or inferred, was present in seventy-six percent. Tetralogy of Fallot occurred in twenty-three patients, about twelve percent, eight of them with pulmonary atresia. Right-sided lesions dominated by a wide margin.

Peripheral pulmonary stenosis is common in Alagille syndrome and uncommon in other causes of infant cholestasis. A review in the European Journal of Human Genetics by Turnpenny and Ellard in 2012 made the point that it is the more specific cardiac finding and should always prompt consideration of the diagnosis. Combined with the Emerick survival data, the practical rule is clear: every suspected case needs an echocardiogram, and the right ventricular pressure must be known before any liver transplant.

Part six. The face

The characteristic face of Alagille syndrome, often described as a broad forehead, deep-set eyes, a straight nose with a bulbous tip and a small pointed chin, has always been controversial. In 1983 Sokol and colleagues argued that a similar face is seen in other forms of chronic cholestasis, and so might reflect the illness rather than the gene. Kamath and colleagues tested this directly in 2002. Clinical dysmorphologists, blinded to the diagnoses, looked at photographs of patients with Alagille syndrome and of patients with other intrahepatic cholestasis. They classified the faces correctly seventy-nine percent of the time. The sensitivity of the facial gestalt was seventy-six percent and the specificity eighty-two percent, with a positive predictive value of eighty-one percent and a negative predictive value of seventy-seven percent. Adult faces were the hardest, because the face coarsens and the chin becomes more prominent with age.

So the face is specific enough to be useful, but its reported frequency ranges widely, from about twenty to about ninety percent according to GeneReviews, depending on who is looking, at what age, and in which ethnic group. In the GALA cohort the face was present in ninety point one percent of patients with JAG1 variants but only fifty-seven point six percent of those with NOTCH2 variants. The face is best treated as a supporting feature, and it is good practice to photograph it for the record.

Part seven. The eye and the spine

These are the two classic features that healthy people also have. Posterior embryotoxon is found on slit-lamp examination in seventy-eight to eighty-nine percent of people with Alagille syndrome. But it is also found in eight to fifteen percent of the general population, and in nearly seventy percent of people with the twenty-two q eleven point two deletion syndrome. Other eye findings are common and useful: optic disc drusen, small deposits at the head of the optic nerve, are seen on ocular ultrasound in about ninety percent, and retinal pigmentary changes in about a

third. Vision is usually preserved.

Butterfly vertebrae are reported in anywhere from thirty-three to ninety-three percent of patients across series, and they are easily seen on a frontal chest or spine radiograph. Vertebral segmentation defects in general occur in many syndromes; the symmetric butterfly form outside Alagille syndrome is mainly seen in the twenty-two q eleven deletion syndrome.

The practical consequence is that a posterior embryotoxon in a well relative is weak evidence on its own, because roughly one healthy person in ten has one. These features are most helpful when they cluster with the heart and the liver. They are also the features that NOTCH2 carriers most often lack: in the GALA cohort, embryotoxon was found in only eighteen point five percent of NOTCH2 patients, and butterfly vertebrae in only three point three percent.

Part eight. The kidney, the forgotten sixth criterion

Kamath and colleagues reviewed four hundred and sixty-six patients with JAG1 variants, of whom one hundred and eighty-seven had enough renal data to evaluate. Thirty-nine percent, seventy-three patients, had a renal anomaly or renal disease. Among those seventy-three, renal dysplasia was by far the most common finding, in fifty-nine percent. Renal tubular acidosis was found in about ten percent, vesicoureteric reflux in about eight percent, and urinary tract obstruction in about eight percent. In the GALA cohort, renal anomalies occurred in thirty-four point five percent of NOTCH2 patients and thirty-nine point two percent of JAG1 patients, a difference that was not significant. In other words, the kidney is spared from the drop-off in classic features that is seen with NOTCH2.

The authors argued that renal involvement should be considered the sixth defining criterion for Alagille syndrome, and that a targeted renal evaluation, with serum biochemistry including creatinine, electrolytes and bicarbonate, a urinalysis and a renal ultrasound, should be the standard of care for every patient. This matters beyond diagnosis. A child who goes on to liver transplantation will receive calcineurin inhibitors, which are toxic to the kidney, and it is far better to know in advance that the kidney is dysplastic.

Part nine. The vessels, a third of deaths

The vascular disease of Alagille syndrome is the feature that kills without warning. Kamath and colleagues reviewed two hundred and sixty-eight patients and found that nine percent had a non-cardiac vascular anomaly or event. The lesions included aneurysms of the basilar, internal carotid and middle cerebral arteries, moyamoya arteriopathy, aortic aneurysms and coarctation, and renal artery stenosis. Most striking, ten of the twenty-nine deaths in the cohort, thirty-four percent, were attributed to vascular causes, mainly intracranial bleeding, some of it after trivial head trauma.

Screening studies show that the disease is more common than symptoms suggest. Emerick and colleagues performed magnetic resonance imaging with angiography, known as MRA, in

twenty-six patients and found cerebrovascular abnormalities in ten, thirty-eight percent, most often stenosis of the internal carotid arteries. Among twenty-two patients screened without symptoms, five, twenty-three percent, had a lesion, and all four symptomatic patients did. The lesions progressed over time in four patients. Importantly, screening missed lesions in two asymptomatic patients who later died of ischaemic events. A later series from Cincinnati by Carpenter and colleagues found cerebral arterial disease, including dolichoectasia, aneurysms and moyamoya, in thirty-two percent of nineteen patients who had dedicated vascular imaging, and venous anomalies in twenty-one percent.

The practical approach recommended in reviews, for example by Cheng and Rosenthal in Hepatology Communications in 2023, is to screen every child with brain MRI and MRA once it can be done without general anaesthesia, and to image urgently after any head trauma or any new neurological sign. A normal study is reassuring but not permanent. Remarkably, no society guideline grades this recommendation.

Part ten. The criteria compared

It helps to set the definitions side by side. The classic definition from Alagille in 1987 requires bile duct paucity on biopsy plus three of five features: cholestasis, heart, vertebrae, eye and face. It fails in infants before six months, in relatives and in NOTCH2 disease. The expanded clinical definition allows cholestasis to stand in for biopsy and counts three of seven features, adding kidney and vascular anomalies, but it still misses mildly affected carriers. GeneReviews adds a family-history route: a person with one of the five major features and a family history of Alagille syndrome meets clinical criteria, which of course depends on the diagnosis in the index patient being secure. GeneReviews also defines a molecular diagnosis, which is established in a person with suggestive findings and a pathogenic or likely pathogenic variant in JAG1 or NOTCH2. A variant of uncertain significance neither establishes nor excludes the diagnosis. Finally, a genotype-anchored definition summarised by Turnpenny and Ellard in 2012 requires a variant plus either a family history or at least one feature, including renal or cardiovascular involvement. Its limitation is the asymptomatic carrier, who has carrier status rather than a syndrome.

The rule to remember is that once the variant is known, the threshold falls to one feature. The three-of-five bar exists to prevent overdiagnosis in strangers who might have some other cause of cholestasis. It should not be used to deny the diagnosis to a child whose genotype has already answered the question.

Part eleven. Criteria in relatives and in adults

The family study by Kamath and colleagues in 2003 examined fifty-three relatives who carried the JAG1 variant found in their proband. Only twenty-one percent, eleven relatives, had features that would have led to the diagnosis on their own. Another thirty-two percent, seventeen relatives, had mild features found only on targeted examination. Forty-seven percent,

twenty-five relatives, did not meet the classic clinical criteria at all, and two had no feature of any kind. Penetrance in this family series was about ninety-six percent, meaning almost every carrier had something, but expressivity varied enormously, so that a parent with only a childhood murmur could share a variant with a child who needs a transplant.

Adults show the same problem. A Chinese series by Li and colleagues in 2023 described eight adults, half with NOTCH2 variants, and only one met the classic criteria. Adult-onset patients had fewer faces, murmurs and embryotoxa, while butterfly vertebrae persisted.

The consequence is simple. Classic criteria describe probands. In relatives, the variant test is the criterion, and the examination measures severity. Above all, classic criteria must never be used to clear a related living liver donor.

Part twelve. NOTCH2 disease fails a JAG1-shaped rule

The classic criteria were drawn from patients who, we now know, almost all had JAG1 variants. The GALA cohort compared nine hundred and two JAG1 patients with thirty-four NOTCH2 patients. A characteristic face was present in ninety point one percent of JAG1 patients versus fifty-seven point six percent of NOTCH2 patients. A cardiac anomaly on echocardiography was present in ninety-two point two percent versus sixty-four point three percent. Posterior embryotoxon was present in fifty-two point eight percent versus eighteen point five percent, and butterfly vertebrae in forty-four point five percent versus three point three percent. All four of those differences were highly significant. By contrast, renal anomalies, neonatal cholestasis and paucity on the first biopsy were similar between the two groups: neonatal cholestasis was actually reported in ninety point six percent of NOTCH2 patients against eighty-three point one percent of JAG1 patients.

So a cholestatic infant with a normal face, eye and spine can still have Alagille syndrome, and the gene panel must include both genes.

Part thirteen. From suspicion to a molecular diagnosis

The yield of genetic testing falls as the number of clinical features falls, but it never reaches zero. In a diagnostic laboratory series summarised by Turnpenny and Ellard, a JAG1 variant was found in fifty-four percent of probands with three features. In a series from Exeter by Guegan and colleagues in 2012, about one third of patients referred with only one or two features carried a JAG1 variant.

The panel matters as much as the decision to test. It should include both JAG1 and NOTCH2, and it must detect copy number changes, because whole-gene deletions account for about thirteen percent of JAG1 variants and are invisible to sequencing alone. If suspicion remains strong after a negative panel, the next steps are a targeted deletion assay such as multiplex ligation-dependent probe amplification, or a chromosomal microarray, and then exome or genome sequencing with later reanalysis. If a variant of uncertain significance is found, testing

the parents to see whether the variant tracks with features in the family, a careful search for other features, and, as the European guideline notes, sometimes a liver biopsy, can help.

The European guideline recommends genetic testing early in the evaluation of unexplained cholestasis in infants and children, using next-generation sequencing rather than single-gene Sanger sequencing. A low threshold for testing finds the mildly affected patients that criteria miss, and every positive result lowers the clinical bar for the relatives who follow.

Part fourteen. The work-up: seven systems, not five

Once Alagille syndrome is suspected or confirmed, the baseline evaluation covers seven systems. For the liver, measure bilirubin, gamma-glutamyl transferase, serum bile acids, lipids, the international normalised ratio and the fat-soluble vitamins A, D, E and K, at diagnosis and then serially, to judge severity, cholesterol and xanthomas, and vitamin deficiency. For the heart, obtain an echocardiogram and a cardiology review, looking for branch pulmonary artery stenosis, tetralogy of Fallot and right ventricular pressure, at diagnosis and again before any transplant. For the spine, a frontal chest or spine radiograph shows butterfly or hemivertebrae. For the eye, a slit-lamp examination looks for posterior embryotoxon, and ocular ultrasound can show optic disc drusen. For the kidney, check creatinine, electrolytes and bicarbonate, a urinalysis and a renal ultrasound, looking for dysplasia, tubular acidosis, reflux and obstruction, at diagnosis and again before calcineurin inhibitors are started. For the brain vessels, perform MRI with MRA when it is feasible without anaesthesia, and urgently after head trauma or any new neurological sign. And for the genes, send a panel that calls copy number before any invasive procedure, and test the parents to establish inheritance.

The work-up is the same whether the diagnosis came from clinical criteria or from a gene. The genotype does not tell you which organs are affected.

Part fifteen. What the guidelines say

There is no dedicated guidance on Alagille syndrome from the American Association for the Study of Liver Diseases. The 2017 joint guideline from the North American and European paediatric societies covers Alagille syndrome only as part of the differential diagnosis of cholestasis in infants. It gives no formal diagnostic criteria, includes gene panels within the cholestasis work-up, keeps liver biopsy as part of that work-up, and advises looking for features of Alagille syndrome before an intraoperative cholangiogram. The 2024 European guideline takes a genotype-based view, noting that most patients carry damaging JAG1 variants and a small minority NOTCH2 variants, recommends early next-generation sequencing, and advises against routine biopsy for prognosis or management while allowing it for a variant of uncertain significance.

Putting these together, the documents differ on the definition of the disease and on the role of biopsy, the North American and European paediatric document being biopsy-inclusive and the European adult document genotype-first. They are aligned on early genetic testing and, in effect,

on avoiding a cholangiogram before the features have been sought. They leave a gap on renal evaluation, which no guideline grades. And they leave a material gap on neurovascular screening: vascular events cause about a third of deaths, yet no society grades MRA screening. The modified criteria for relatives, too, live only in reviews and in GeneReviews.

Part sixteen. Common traps

Several errors recur. The first is excluding Alagille syndrome because a biopsy before six months shows ducts, when paucity is present in only sixty percent of early biopsies. The second is reading absent isotope excretion as biliary atresia, when sixty-one percent of infants with Alagille syndrome do not excrete by twenty-four hours. The third is proceeding to a Kasai without an echocardiogram, a spine film, an eye examination and a gene panel, given that transplant rates were about forty-seven percent versus fourteen percent and death rates about thirty-two percent versus three percent after a Kasai in the Philadelphia series. The fourth is demanding three of five features once a pathogenic variant is known, when one feature is enough with a variant or a family history. The fifth is clearing a related donor because the donor does not meet criteria, when forty-seven percent of carrier relatives do not. The sixth is counting a lone embryotoxon in a healthy relative as proof, when eight to fifteen percent of the general population has one. The seventh is expecting the classic face, eye and spine in NOTCH2 disease, where butterfly vertebrae occur in about three percent and embryotoxon in about nineteen percent. The eighth is stopping at five systems, when renal disease affects thirty-nine percent and vascular events cause a third of deaths. The ninth is treating a normal screening MRA as permanent reassurance, when lesions progress and normal studies have preceded fatal events. And the tenth is relying on the facial gestalt in infants or adults as a decisive criterion, when it is least reliable at both ends of life.

Part seventeen. What to do differently in clinic

Return to the infant at the start of this chapter. She has two of the five classic features, cholestasis and branch pulmonary stenosis, and no paucity on a biopsy taken at seven weeks. That combination proves nothing either way at her age. The correct response is to postpone the cholangiogram and the Kasai, obtain a frontal spine film and a slit-lamp examination, and send a cholestasis panel that includes JAG1 and NOTCH2 and detects deletions. If a pathogenic JAG1 variant is found, the diagnosis is made, because a variant plus any feature is sufficient. The work-up then extends to the kidney, with creatinine, bicarbonate, urinalysis and ultrasound, and later to the brain vessels with a non-sedated MRI and MRA. Her mother is tested for the familial variant; she is not diagnosed or excluded on the basis of her eye, because about one healthy person in ten has an embryotoxon.

More broadly, four changes follow from this chapter. Think of Alagille syndrome in every infant with high-GGT cholestasis and a murmur, and seek the features before any operation on the bile ducts. Use the classic rule only for patients without a genetic answer, and once the variant is known, accept one feature. Examine seven systems rather than five, because the kidney and the

vessels carry much of the hidden morbidity. And use the genotype, not the examination, to decide who in the family is affected, especially when a relative offers to become a living liver donor.

The criteria describe probands. The infant liver hides its paucity. The kidney and the vessels cause harm quietly. And the variant lowers the bar to a single feature.